LEVEL 1: Master Unit Testing Registry

Series 100: User Authentication, Security & Session Management

Focuses on crypto hashes, formatting rules, token lifetimes, and gatekeeping rules.

Unit ID	Target Function / Code Method	Testing Scenario Profile	Expected Isolation Output
UT-101	validateEmailFormat(email)	Input contains structural typos during Patient Registration (e.g., test@com, @@domain.com).	Return false (Rejected)
UT-102	validatePasswordStrength(pwd)	Input is less than 8 characters or misses a special character.	Return false (Weak password)
UT-103	hashUserPassword(plainText)	Raw string is passed into the cryptographic library (e.g., BCrypt) to secure patient data.	Return a unique 60+ char secure salt hash.
UT-104	verifyPasswordMatch(plain, hash)	Matching a user login password attempt against database hash record.	Return true if match; false if modified.
UT-105	generateSessionToken(userId)	User authenticates successfully; token needs generation for RBAC assignment.	Return structured JWT or secure random state string.
UT-106	isTokenExpired(timestamp)	Token expiry timestamp is older than current server system time.	Return true (Trigger auto-logout routine)
UT-107	sanitizeHtmlInput(userInput)	Input contains unsafe script tags (e.g., <script>malicious()</script>). in the medical history box	Return stripped or encoded clean string (XSS protection).

LEVEL 2: Master Integration Testing Registry
Series 100: Frontend Interface < ------- > Backend API Integration

Validates the data-sharing bridge between user input screens and server-side processing controllers.

Integration ID	Interoperating Subsystems	Testing Scenario Profile	Expected Cross-Module Integration Behavior
IT-101	UI Registration Form < —> Auth Controller	A user inputs registration details on the screen and clicks "Submit".	The frontend gathers the form array, serializes it to JSON, dispatches an HTTP POST request, handles the server's HTTP 201 Created status code, and redirects to login.
IT-102	Login Screen < —> State/Session Manager	A user logs into an authenticated account profile.	The backend API returns a session token; the client-side framework catches it, securely commits it to storage (localStorage or cookies), and updates global application access states (Admin/Secretary/Patient).
IT-103	Interactive Odontogram UI < —> Chart Controller	The dentist clicks a specific tooth graphic to log a cavity or filling.	The frontend captures the click event, maps it to the specific tooth number (Adult/Pediatric), fires an API query, and dynamically updates the visual chart colors without refreshing the web view.
IT-104	Booking Calendar UI < —> Schedule Controller	A patient selects a dental service and attempts to book a date.	The client app sends the procedure type; the backend calculates the required time duration, checks the database for conflicts, and returns only the available time blocks to the calendar UI.

LEVEL 3: Master System Testing Registry

Series 100: User Authentication, Security & Account Management

Ensures that access gates work perfectly and user accounts are managed securely.

System Test ID	Module	Step-by-Step Execution Procedure	Test Data / Inputs	Expected System Response
ST-101	Account Registration	1. Navigate to the Registration page. 2. Fill fields using an email format that already exists in database. 3. Click "Register".	Email: existing_user@gmail.com Password: Wrong Pass123	Form submission is blocked. A clear warning message states, "Email is already registered".
ST-102	Login Security	1. Navigate to Login page. 2. Enter a valid username but an incorrect password. 3. Click "Login".	Password: WrongPass123	Login fails. System displays an error message: "Invalid credentials." Password field is wiped clean.
ST-103	Password Masking	1. Navigate to Login page. 2. Type values into the password field.	Any input string	Input values are visually masked using bullet dots (••••••) to prevent shoulder surfing.

ST-104	Form Validation	1. Navigate to Registration page. 2. Leave required fields blank. 3. Click "Submit".	Blank fields	Form submission fails. Inline error messages label each blank field with "This field is required."
ST-105	Role-Based Access	1. Log in using a standard Patient account profile. 2. Try to directly type the Admin dashboard URL link into the browser tab.	URL: /admin/dashboard	System blocks entry, terminates or redirects the request, and displays an "Access Denied" page.
ST-106	Session Logout	1. Log into an active account session. 2. Click the "Logout" button. 3. Use the browser's "Back" arrow button.	Click navigation arrows	System prevents the user from accessing the dashboard again and redirects them to the login screen.

Series 200: Core Data Management & Transaction Processing (CRUD)

Validates the system's ability to smoothly Create, Read, Update, and Delete information.

System Test ID	Module	Step-by-Step Execution Procedure	Test Data / Inputs	Expected System Response
ST-201	Record Creation	1. Navigate to the core creation form [e.g., Book Appointment] 2. Populate valid data fields. 3. Click "Save".	Valid record values	A success popup alert confirms saving. The system automatically redirects the user back to the master list view.
ST-202	Input Data Boundaries	1. Open the data entry form. 2. Type text characters into fields explicitly meant for numbers. 3. Click "Save".	Stock Quantity: "Five"	System rejects input, stops processing, and displays a validation alert: "Please enter a valid numeric value."
ST-203	Record Modification	1. Select an existing entry row from the data table. 2. Click "Edit", modify a field value, and click "Update".	Updated name/value string	Data grid instantly updates with the new modifications. Changes are successfully preserved upon page refresh.
ST-204	Record Removal	1. Select a record entry row from the data table. 2. Click the "Delete" trash icon.	Click action button	A modal popup asks: "Are you sure you want to delete this record?" Data is dropped only after confirmation.
ST-205	Table Pagination	1. Navigate to a heavy data grid list containing more entries than the display limit *[e.g., 50 rows total, limit 10]*.	Click "Next" / "Page 2"	The table accurately displays the next set of records while keeping the previous records preserved in memory.

		2. Interact with footer page controls.		

Series 300: Search, Filtering & Information Retrieval

Ensures users can efficiently query, sort, and look up information in the database.

System Test ID	Module	Step-by-Step Execution Procedure	Test Data / Inputs	Expected System Response
ST-301	Global Text Search	1. Open a master list registry view. 2. Type a known, specific keyword identifier into the search input box.	Keyword query	Data grid filters rows instantly, displaying only records that contain the specified search keyword string.
ST-302	Empty Search State	1. Type a random, non-existent keyword string into the search input box.	Query: "xyz987abc"	The data table clears out its row views and displays a clear message card stating: "No matching records found."
ST-303	Multi-Filter Sorting	1. Open the data tracking grid panel. 2. Select filter criteria properties *[e.g., Status: Active, Category: Electronics]* and click apply.	Drop-down selection filters	Table hides irrelevant entries and groups matching rows accurately according to the active filters selected.

Series 400: Reporting, Document Exports & Analytic Computations

mathematical reliability and the generation of business reporting files.

System Test ID	Module	Step-by-Step Execution Procedure	Test Data / Inputs	Expected System Response
ST-401	Mathematical Audit	1. Run transactions with manual reference calculations (e.g., Billing module). 2. Open the system's financial analytics view.	Simulated record data	System-calculated values (sums, averages, taxes) exactly match the verified manual calculation totals.
ST-402	PDF Document Export	1. Navigate to a reporting table summary screen. 2. Click the "Export to PDF" download button.	Click action button	System compiles the current data and initiates a browser download of a clean, structured .pdf file asset.
ST-403	Spreadsheet Export	1. Navigate to a reporting table summary screen. 2. Click the "Export to Excel/CSV" download button.	Click action button	System cleanly formats records into comma-separated rows and downloads an uncorrupted Excel-readable file.

Series 500: System Configurations & Notifications

Tests backend automation, profile modifications, and interface status alert behaviors.

System Test ID	Module	Step-by-Step Execution Procedure	Test Data / Inputs	Expected System Response
ST-501	UI Feedback Alerts	1. Trigger a background process activity [e.g., Sending Emails] via form action.	Click trigger switch	A loading animation display blocks the UI screen temporarily, preventing duplicate form submissions from rapid clicks.
ST-502	Settings Profiling	1. Navigate to user settings or general options panel. 2. Change system configurations and save.	Update configuration choices	The system successfully stores the new configurations and they immediately reflect across the relevant modules.

LEVEL 4: COMPREHENSIVE USER ACCEPTANCE TESTING (UAT) SURVEY

PART 1: PARTICIPANT PROFILE MATRIX
Students must document exactly who tested the system. A minimum of 3 to 5 domain experts/clients and 10 to 15 end-users is standard for an IT Capstone.

Participant ID	Target User Role / Position	Organization / Department	Testing Date	Signature / Initials
UAT-EX-01	Administrator (Expert)	Clinic Administration	07/15/2026	*Signed*
UAT-EX-02	Dentist (Expert)	Azucena Dental Clinic	07/15/2026	*Signed*
UAT-EU-01	Secretary (End-User)	Operations Dept.	07/16/2026	*Signed*
UAT-EU-02	Patient (End-User)	External Stakeholder	07/16/2026	*Signed*

PART 2: FULL ISO/IEC 25010-ALIGNED SURVEY EVALUATION
This instrument evaluates the system using the industry standard **ISO/IEC 25010 Software Quality Model**.

Likert Rating Scale:

5 – Strongly Agree (SA) | 4 – Agree (A) | 3 – Neutral (N) | 2 – Disagree (D) | 1 – Strongly Disagree (SD)

1. Functional Suitability

Measures how well the software solves the actual business problems it was built for.

#	Evaluation Criteria Indicators	5	4	3	2	1
1.1	**Functional Completeness:** The system provides all required features and modules outlined in the project objectives.					
1.2	**Functional Correctness:** The system handles data entries accurately and computes results/reports without mathematical errors.					
1.3	**Functional Appropriateness:** The workflows programmed into the software align properly with real-world business policies.					

2. Usability & User Experience (UX)

Evaluates how easy, intuitive, and stress-free the system is for the target users.

#	Evaluation Criteria Indicators	5	4	3	2	1
2.1	**Appropriateness Recognizability:** Users can quickly understand what each module, button, and menu does upon first glance.					
2.2	**Learnability:** The system is easy to learn; new users can perform core operations without intensive training or manuals.					
2.3	**Operability:** Interface components (forms, search inputs, filters) are easy to control and interact with on the screen.					
2.4	**User Interface Aesthetics:** The color schemes, typography, layout grid, and overall graphic design are professional and visually clean.					
2.5	**Accessibility:** The text font sizes, icons, and interactive elements are highly visible and accessible to diverse users.					

3. Performance Efficiency

Tracks how well the system utilizes server resources, bandwidth, and user time.

#	Evaluation Criteria Indicators	5	4	3	2	1
3.1	**Time Behavior:** Processing times, page transitions, database queries, and report generation execute within acceptable speeds.					
3.2	**Resource Utilization:** The system runs smoothly without causing browser lag, application hangs, or massive device memory consumption.					

4. Security

Ensures that sensitive data remains locked down and protected from unauthorized adjustments.

#	Evaluation Criteria Indicators	5	4	3	2	1
4.1	**Confidentiality:** The software ensures data is visible only to authorized personnel (e.g., standard users cannot access admin portals).					
4.2	**Integrity:** The system prevents unauthorized modification of database files, tracking record updates securely.					
4.3	**Accountability:** Actions taken within the software can be accurately traced back to the specific logged-in user account.					

5. Reliability

Validates that the software can maintain its level of performance over extended usage.

#	Evaluation Criteria Indicators	5	4	3	2	1
5.1	**Fault Tolerance:** The application handles human entry mistakes gracefully without crashing the server or throwing raw code errors.					
5.2	**Recoverability:** In the event of a network interruption or failure, the system can preserve inputs or recover its operational state safely.					

6. Maintainability & Portability

Assesses how well the system runs across different tech ecosystems and how easy it is to update later.

#	Evaluation Criteria Indicators	5	4	3	2	1
6.1	**Co-existence / Compatibility:** The software runs correctly across multiple browsers (Chrome, Edge, Safari) and screen sizes without breaking components.					
6.2	**Modularity / Reusability:** The application components are structured cleanly so that future developers can easily add new updates or sub-features.					

PART 3: FORMAL CLIENT ACCEPTANCE & SIGN-OFF
This signature sheet is the definitive evidence required by capstone panels. It proves the capstone project solved a real institutional need and has been officially handed over.

Acceptance Statement:

"By signing below, I acknowledge that I have actively participated in or reviewed the User Acceptance Testing (UAT) phase for the developed system. I confirm that the functional deliverables, user interfaces, security protocols, and operational workflows meet our agreed specifications and are accepted for live deployment."

Approved By:

Client Printed Name & Designation

Signature over Printed Name

Date Signed: _____ / _____ / 2026